

Optimal Strategy in Shut the Box

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Abstract

Shut the Box could be considered a game of luck. Roll two dice, flip tiles, and repeat until you get stuck. But on any given turn, there are often three or four valid combinations to choose from, and picking the wrong one quietly kills your chances. This paper determines the optimal decision for each roll from 2 to 12, validated across one million simulations.

1 Introduction

Nine tiles, numbered 1 through 9, start face-up. You roll two dice and flip down any combination of open tiles that sum to your roll. You repeat until no move exists. Your score is the sum of tiles face-up when no moves exist. A win is a score of zero. For example, a roll of 6 can be satisfied by:

$$\{6\}, \quad \{1,5\}, \quad \{2,4\}, \quad \{1,2,3\}.$$

Four valid moves, four different boards left behind.

Most people who play go by feel. Roll a 7, flip the 7. Roll a 9 when the 9 is gone, and flip whatever adds up. However, there is a right answer.

2 Model

Consider the set of face-up tiles as $S \subseteq \{1, 2, \dots, 9\}$. Each tile is up or down, giving at most $2^9 = 512$ board configurations, although most of them cannot actually appear in a real game.

Two dice are rolled each turn. The probability of rolling both die and getting $r \in \{2, \dots, 12\}$ is

$$P(r) = \frac{6 - |r - 7|}{36}.$$

Seven comes up most often at $6/36$. Two and 12 each come up $1/36$ of the time.

Given board S and roll r , write $A(S, r)$ for the collection of subsets of S adding up to r . No valid subsets means the game ends with a score of $\sum_{t \in S} t$. Otherwise, the player picks some $C \in A(S, r)$ and the board becomes $S \setminus C$.

We look at two versions of the problem. MaxWin finds the strategy that gives the best shot at scoring zero. MinLoss finds the strategy that keeps the final score as low as possible. Close, but not the same thing.

3 Analysis

3.1 Solving the Game

Both versions are solved by starting at the end of the game and working backwards. For MaxWin, $W(S)$ is the probability of winning from board S under optimal play:

$$W(\emptyset) = 1$$

$$W(S) = \sum_{r=2}^{12} P(r) \cdot \max_{C \in A(S, r)} W(S \setminus C)$$

The max is zero when $A(S, r)$ is empty. At each board and roll, the right move is whichever C gives the highest $W(S \setminus C)$.

For MinLoss, $E(S)$ is the expected final score from board S :

$$E(\emptyset) = 0$$

$$E(S) = \sum_{r=2}^{12} P(r) \cdot \min_{C \in A(S, r)} E(S \setminus C)$$

When $A(S, r)$ is empty, the min is just $\sum_{t \in S} t$.

Since every move takes at least one tile off the board, this always finishes. The numbers come out to $W(\{1, \dots, 9\}) \approx 7.14\%$ and $E(\{1, \dots, 9\}) \approx 11.16$. Playing perfectly still means losing more than nine times out of ten, and walking away with about 11 points left on average. Not a forgiving game.

3.2 Which Tile to Flip

Table 1 shows the right move for every opening roll, and what to do if the same roll comes up again right after.

Table 1: Optimal tile choice for each opening roll, and optimal choice if the same roll occurs again.

Roll	First Roll	Same Roll Again
2	{2}	–
3	{3}	{1, 2}
4	{4}	{1, 3}
5	{5}	{2, 3}
6	{6}	{2, 4}
7	{7}	{2, 5}
8	{8}	{3, 5}
9	{9}	{1, 8}
10	{1, 9}	{2, 8}
11	{2, 9}	{3, 8}
12	{3, 9}	{5, 7}

For rolls 2 through 9, the right move is always:

$$C^* = \{r\}.$$

Just flip the tile with that number on it. For rolls 10 through 12, no single tile exists, so include the 9 and pair it with whatever fills the gap.

Roll 2 is a forced move with no decision, and rolling it a second time ends the game immediately since tile 2 is already gone. Roll 3 comes down to {3} versus {1, 2}. Dropping both the 1 and 2 at once is a steep price, so {3} is the call.

The reason the single tile rule works is that small tiles are the most valuable pieces on the board. The 1, 2, and 3 pair with nearly every possible roll, so losing them early leaves you stuck. A single large tile comes off clean, and everything else stays put.

When the same roll comes up a second time, the single tile is gone, and you are choosing from two-tile combinations. Most rolls: avoid the small tiles, keep them for later. Roll 9 is the exception. The right second move is {1, 8} rather than {2, 7}, {3, 6}, or {4, 5}, because getting rid of the 8 early is worth losing the 1.

3.3 Does the Objective Matter?

Across 1,000,000 games with shared dice rolls, MaxWin and MinLoss barely differ. They only split on late-game boards with almost no tiles left, which do not come up often enough to move the overall numbers. NoHigh, which always grabs the highest tile with no lookahead, trails both but not by much. The gap between playing optimally and playing naively turns out to be pretty thin.

Table 2: Simulation results over 1,000,000 games with identical dice rolls across all strategies.

Metric	MaxWin	MinLoss	NoHigh
Average score	11.20	11.16	11.37
Wins	71,239	70,685	69,419
Win rate (%)	7.12	7.07	6.94

4 Conclusion

Flip the single tile that matches your roll. If the roll is above 9, take it and pair with whatever fills the gap. For later turns, preserve the smallest tiles as long as possible and avoid splitting a sum across more tiles than you need to. The two objectives, maximizing wins and minimizing score, sound different but lead to the same decisions almost every time, with a win rate of around 7%.

References

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- [3] Anthropic. *Claude*. <https://claude.ai>